

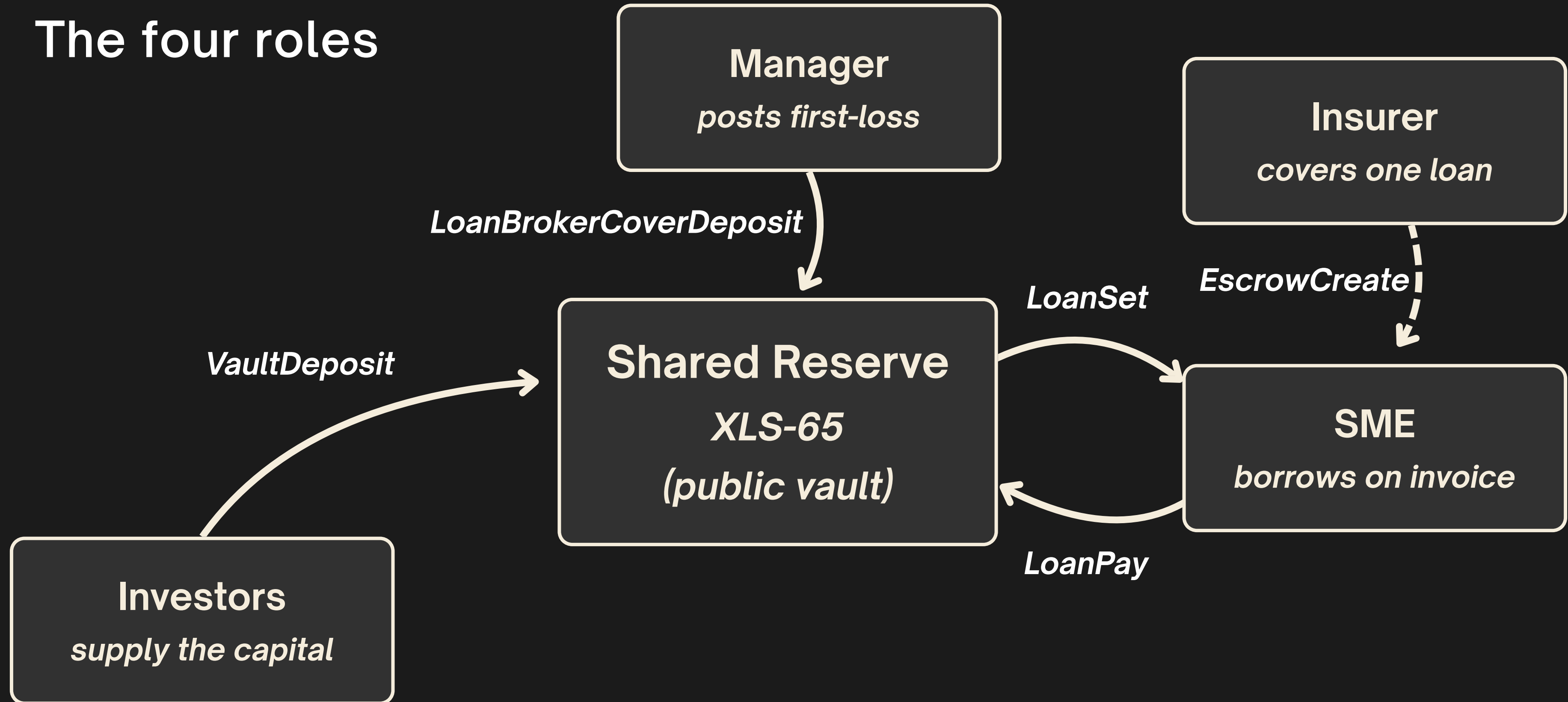
linear flow

Track 1

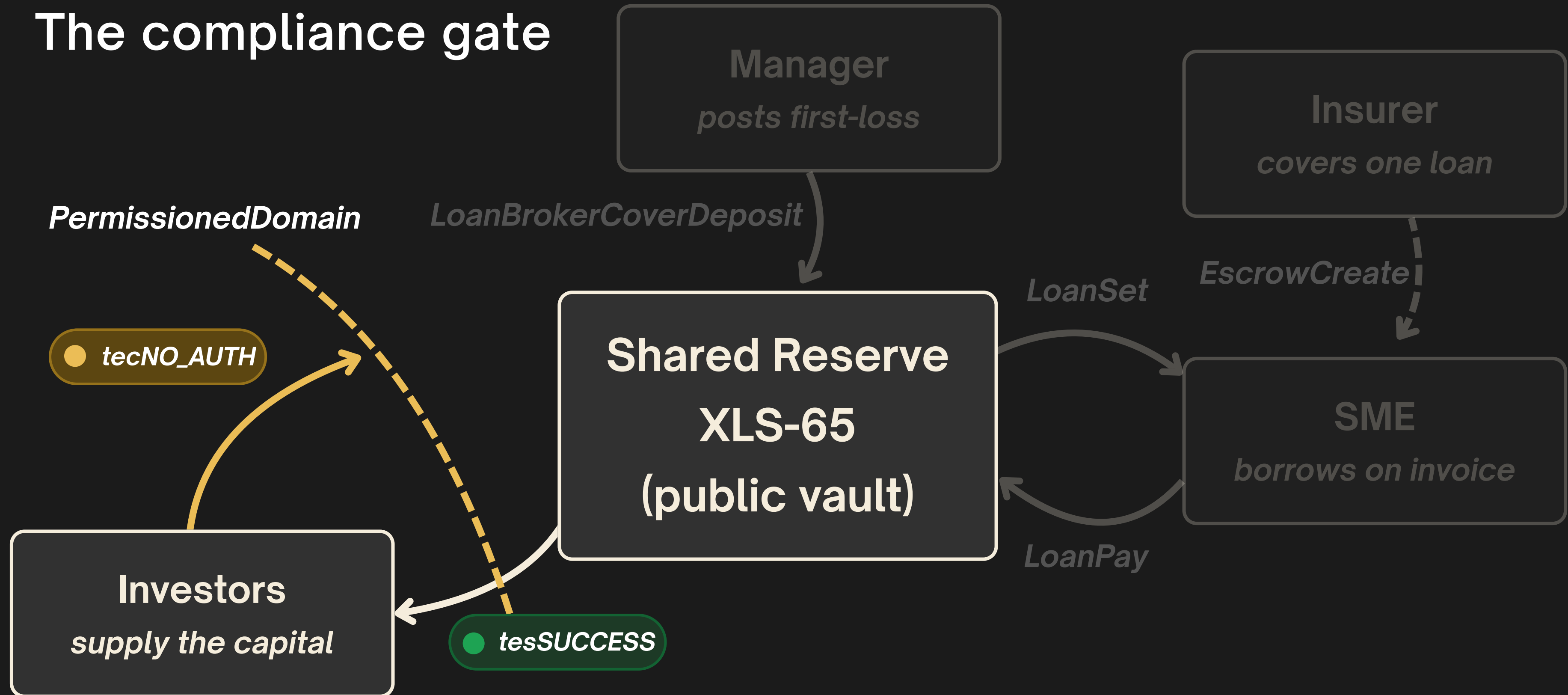
Invoice factoring on
the XRP Ledger

@azelout
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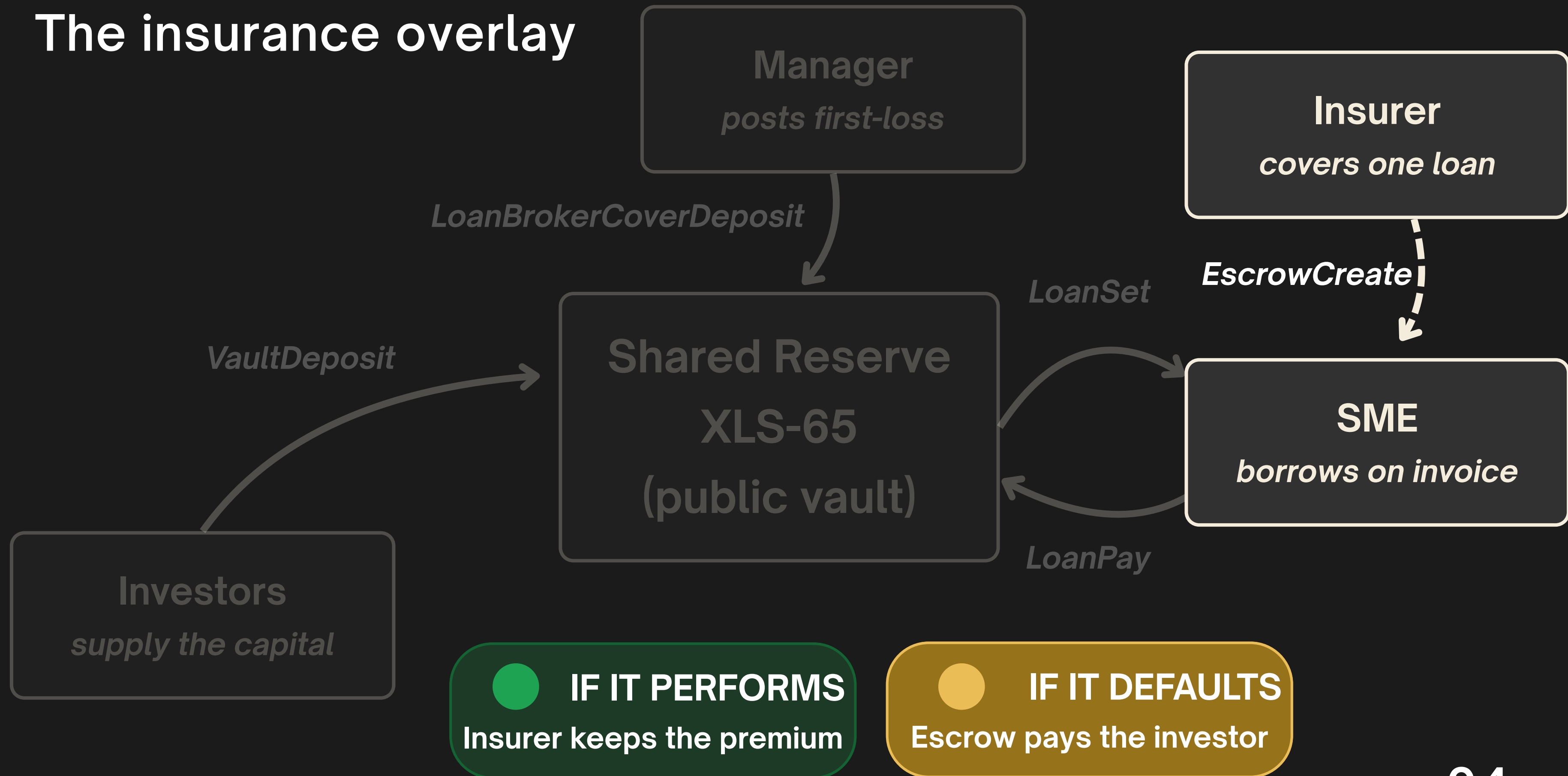
The four roles



The compliance gate



The insurance overlay



PROBLEM 01 // 04

Escrow can't watch a Loan

► *Insurance has to pay on default. Nothing can see the default.*

- *`LoanManage tfLoanDefault`* sets the flag on the Loan
- *`EscrowFinish`* only releases on a time condition or a fulfillment
- No lock in the protocol can read another object's state
- Our workaround: the manager holds the fulfillment, as a named and disclosed trusted party

PROBLEM 02 // 04

No browser wallet signs a VaultDeposit

▶ *Every documented path uses a server-held seed instead*

- Wallet crashes its own popup on ``VaultDeposit``
- WalletConnect silently swaps ``xrpl:4001`` for ``xrpl:0``, so we removed it
- The reference app and the xrpl.org tutorial both sign server-side

PROBLEM 03 // 04

The last installment cannot close

▶ *Paying a loan off is the one payment that fails*

- *`LoanPay tfLoanFullPayment` returns `tecKILLED` on the final installment*
- *Every earlier installment accepts the same flag*
- *No close-out path is documented*

PROPOSED FIX

accept the flag on the final installment, or document the intended path

PROBLEM 04 // 04

A private vault gates deposits, not loans

▶ *The vault refused the deposit, then funded the loan*

- *`XLS-65 §3.5.2.2 #6` refuses `VaultDeposit` from outside the domain*
- *`XLS-66 §3.8.5.2` lists 24 failure conditions for `LoanSet`, and `DomainID` is in none of them*
- *Same account, same ledger state, next transaction*
- *Reproduced identically on two independent runs*

PROPOSED FIX

have LoanSet consult the vault's DomainID, or scope vault privacy in the spec

dual-signed, so broker discretion stops it, not the protocol

*Thanks for
keeping up !*

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Q&A Time

<https://github.com/TPautras/XRPL-Lending-Joutix>